TEAM NEURO-SIGINT

Automatic Modulation Classifier

Machine learning-based signal classification deployed on FPGA hardware for real-time modulation type identification from raw IQ samples

5 Modulation Types

AM, FM, Carrier, Digital, Noise classification

FPGA Deployment

SVM trained in MATLAB, deployed via VHDL

87% Accuracy

Current performance on 6000-sample dataset

System Architecture & Signal Processing Pipeline

Processing Specifications

- 4096-sample frames with fs =200 kHz sampling rate
- SNR range: 8–20 dB for robust classification
- 16-bit ADC simulation for realistic quantization
- Real-time inference capability on FPGA

Signal Generation

Simulated IQ data with various modulation schemes

Feature Extraction

Five hand-crafted statistical features computed

SVM Classifier

Linear SVM with One-vs-All decision logic

FPGA Deployment

Fixed-point quantization and VHDL implementation

Confidence Gate

Detection based gate logic for reliable output

Attach with PDW

Attached the output into the PDW processing chain

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Feature Engineering: Five Discriminative Features

Hand-crafted statistical features derived from instantaneous frequency and envelope measurements effectively discriminate between modulation classes by capturing unique signal characteristics.

1 **Frequency Variance**

Measures frequency deviation stability—AM shows low variance, FM shows high variance

2 **Frequency Peakiness**

Captures spectral concentration—Carrier signals exhibit sharp peaks in frequency distribution

3 **Zero-Crossing Rate**

Counts signal transitions through zero—digital modulations show higher rates than analog

4 **Magnitude Variance**

Quantifies amplitude fluctuations—AM shows high variance, FM and carrier show low variance

5 **Peak-to-Average Power Ratio**

Compares peak to average power—reveals crest factor differences across modulation types

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Performance Results & Enhancement Roadmap

Current Performance

87% accuracy achieved on 6000-sample dataset with 80/20 train-test split. Performance degrades at lower SNR levels, indicating need for robustness improvements.

RBF Kernel SVM

Non-linear kernel for improved class separation

More Modulation Classes

Expand to 8+ classes including QAM, PSK variants

Lower SNR Training

Train down to 0 dB for robust low-SNR operation

Higher-Order Statistics

Include cumulants and time-frequency features

Target: 93%+ accuracy through algorithmic and feature engineering enhancements

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FPGA Deployment & Real-World Applications

01

Model Training in MATLAB

Linear SVM trained on simulated IQ data with 5 modulation classes

02

Weight Export to VHDL

Generate svm_params_pkg.vhd containing fixed-point quantized parameters

03

Fixed-Point Quantization

16-bit weights (Q1.15 format), 48-bit biases for precision preservation

04

One-vs-All Decision Logic

Parallel classifier execution with maximum-score winner selection

05

Real-Time Inference

4096-sample frame processing in microseconds on FPGA hardware

Software Defined Radio

Adaptive demodulation in SDR receivers for flexible communications

Spectrum Monitoring

Regulatory compliance and unauthorized signal detection in RF environments

Cognitive Radio

Dynamic spectrum access and modulation adaptation for efficient utilization

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Why FPGA & SVM for Constrained Devices

FPGA Advantages

Parallel processing: Execute multiple operations simultaneously without sequential bottlenecks

Deterministic latency: Fixed, predictable inference time—critical for real-time signal processing

Low power consumption: No OS overhead, optimized hardware-only computation

Reconfigurability: Adapt to new modulation classes without replacing hardware

No memory bandwidth bottleneck: Direct hardware implementation avoids CPU-memory transfers

SVM vs Neural Networks

Smaller model size: SVM weights fit in kilobytes vs megabytes for NN, crucial for FPGA memory constraints

Fewer computations: Linear SVM requires simple dot products vs matrix multiplications in NN layers

Lower power per inference: Minimal floating-point operations reduce energy consumption

Easier quantization: Fixed-point conversion (Q1.15) maintains accuracy without precision loss

Proven on small datasets: SVM generalizes well with 6000 samples; NN requires millions for comparable performance

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